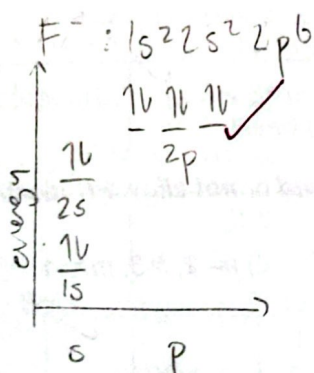


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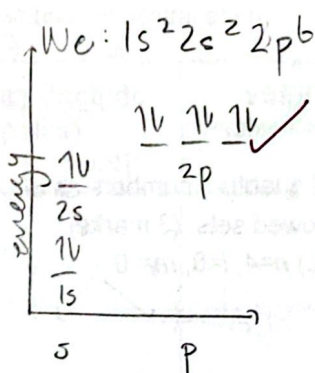
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Chapter 3 Quiz: Atomic Theories**Part A: Multiple Choice (20 marks - on D2L)****Part B: Thinking (12 marks)**

1. Draw the energy level diagrams for the ion of fluorine and the element neon. How do these diagrams compare? (3 marks)



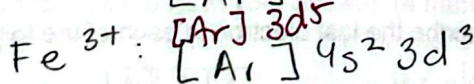
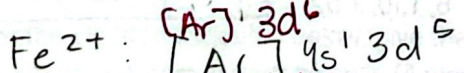
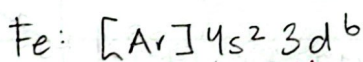
Fluorine



Neon

these energy level diagrams are the same. Fluorine ion gains 1 e^- to become stable, which fills the 2p orbital, having the same # of e^- as neon, so having the same diagram

2. Propose electron configurations (using the noble gas configuration) for both of the ions of iron. Provide a BRIEF explanation. (3 marks)

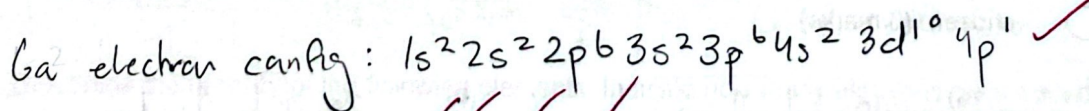


Fe^{3+} has 23 e^- and is not an anomaly, so no e^- are taken from the s orbital

→ Take from highest quantum # first (4s) $(26 - 2 = 24)$
 Fe^{2+} has 24 e^- and is

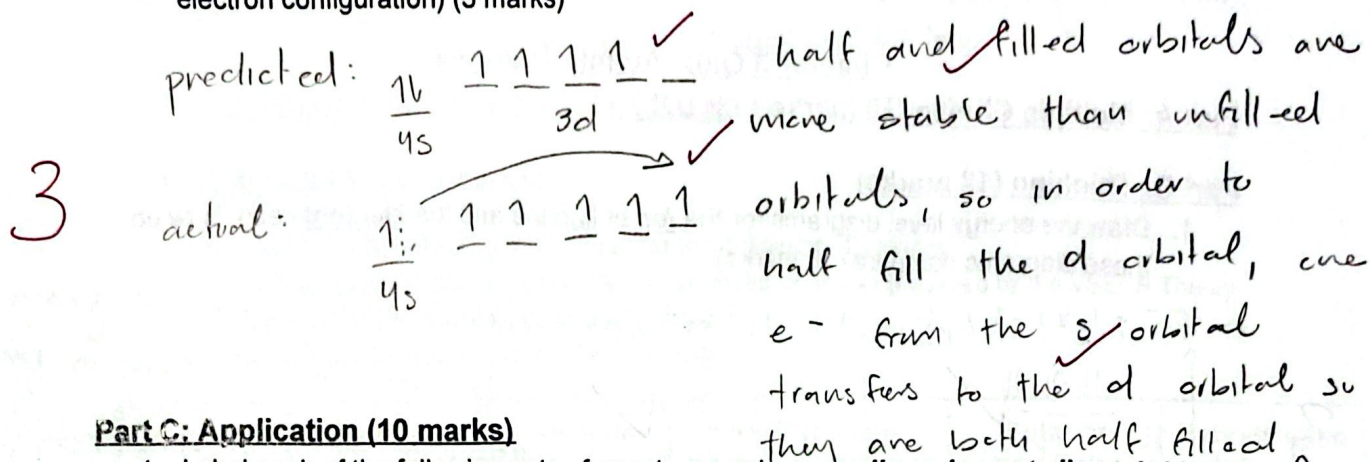
expected to be $[Ar] 4s^2 3d^4$ but one e^- from the s orbital moves to the d orbital because half filled orbitals are more stable than unfilled orbitals.

3. Write the full electron configuration for gallium AND list the set of quantum numbers for the last valence shell electron of this element. (3 marks)



Ga quantum #: $(4, 1, -1, +\frac{1}{2})$

4. Explain chromium's anomalous electron configuration using the valence shell of this element's energy level diagram. (HINT: Show a comparison with chromium's predicted electron configuration) (3 marks)



Part C: Application (10 marks)

1. Label each of the following sets of quantum numbers as **allowed** or **not allowed**. Identify the problem for each of the not allowed sets. (3 marks)
- a) $n=2, l=2, m_l=0$ not allowed ✓
l values cannot equal n values, they must go from 0 to $(n-1)$
- b) $n=4, l=0, m_l=0$ allowed ✓
- c) $n=2, l=3, m_l=-1$ making the not allowed atom more stable
l values cannot exceed n values, they must range from 0 to $(n-1)$
2. Identify the element whose highest energy electron would have the following four quantum numbers: (2 marks)

2

3, 2, +1, +1/2 Cr (Chromium) 6, 1, 0, +1/2 Pb (Lead)

3. Write the four quantum numbers which describe the last electron in each of the following: (2 marks)

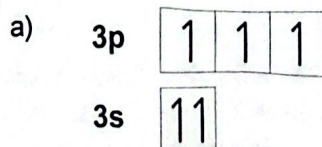
2

Argon $(3, 1, +1, -\frac{1}{2})$ Magnesium $(3, 0, 0, -\frac{1}{2})$

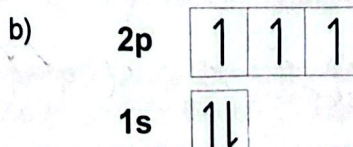
4. Which of the following "rules" (Hund, Pauli Exclusion Principle, Aufbau) is being violated in each orbital diagram below? Provide a **BRIEF** explanation of the rule you have chosen. (3 marks)

Pauli: 3 electrons

may not both have the same spin in a subshell



Pauli's exclusion principle is being violated, as 2 electrons in the same orbital have the same spin (1)



Aufbau's exclusion principle is being violated, as 2p orbital is being filled before the 2s one

Aufbau: lowest energy orbitals must be filled with e^- before higher ones